Zander W. Blasingame

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Summary

I am a postdoctoral fellow working with Alex Tong at Aithyra researching controllable generative models for ai4science applications. My research expertise is in generative modeling (diffusion, flows, and GANs), neural differential equations, stochastic processes, dynamical systems, and biometrics.

Education

Clarkson University

Ph.D. in Electrical and Computer Engineering 2020 - 2025

Thesis: On Guided and Reversible Solvers for Neural Differential Equations

M.Sc. in Electrical and Computer Engineering 2018 - 2025

B.Sc. in Computer Engineering, with Great Distinction 2015 - 2018

Research & Industry Experience

AITHYRA Jan 2026 - Present

Postdoctoral Fellow *Vienna, Austria*

- Researching efficient architectures for *cheap* log density evaluations with CNFs
- Developing a family of neural stochastic flow maps

Clarkson University May 2025 - Dec 2025

Postdoctoral Research Assistant *Potsdam, NY*

- Developed *Rex* a family of algebraically reversible solvers for diffusion models
- Researching controllable schemes for editing real samples for image editing

Clarkson University 2018 - 2025

Graduate Research Assistant *Potsdam, NY*

- Published 15 papers, 8 first author, in highly competitive venues such as NeurIPS ($< 30\%$ acceptance rate)
- Trained large-scale diffusion models using distributed training over multiple GPUs and nodes in a high-performance computing cluster
- Optimized the performance of end-to-end optimization techniques for flow/diffusion models by using a greedy strategy to decrease the runtime complexity per optimization step from $\mathcal{O}(nd^2)$ to $\mathcal{O}(d^2)$
- Created an *algebraically reversible* solver for diffusion SDEs for improved stability in image editing applications and reduced computation overhead by dynamically calculating the Brownian motion
- Developed bespoke numerical solvers for efficiently performing backpropagation through the ODE/SDE solve of diffusion models with a reduction in memory usage from $\mathcal{O}(n)$ to $\mathcal{O}(1)$ whilst reducing the error in estimating the gradients by transforming the ODE into a non-stiff form
- Proposed a new family of face morphing attacks known as **Di**ffusion **M**orphs (**DiM**) which have become the state-of-the-art in face morphing attacks, with effectiveness against face recognition systems
- Developed an efficient GAN architecture for encoding images into the latent space by sharing the CNN backbone for the discriminator and encoder networks
- Designed a novel detection algorithm for face morphing attacks by using adversarial learning to train a better detection scheme

Clarkson University 2016 - 2018

Undergradute Research Assistant *Potsdam, NY*

- Investigated probability density estimation using GAN inversion
- Created a novel GAN model to generate out of distribution data for anomaly detection
- Developed state-of-the-art machine learning algorithms to detect zero-day malware attacks using low-level hardware features

Griffiss Institute - Air Force Research Laboratory Summer 2016
Engineering Intern *Rome, NY*

University of New Hampshire InterOperability Laboratory Summer 2014
Engineering Intern *Durham, NH*

Peer-reviewed Conference & Journal Publications

[1] **Zander W. Blasingame** and Chen Liu. "Greed is Good: A Unifying Perspective on Guided Generation". In: *The Thirty-ninth Annual Conference on Neural Information Processing Systems*. 2025. URL: https://openreview.net/forum?id=s14pdQgoLb.

[2] **Zander W. Blasingame** and Chen Liu. "AdjointDEIS: Efficient Gradients for Diffusion Models". In: *Advances in Neural Information Processing Systems*. Ed. by A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang. Vol. 37. Curran Associates, Inc., 2024, pp. 2449–2483. URL: https://proceedings.neurips.cc/paper_files/paper/2024/file/04badd3b048315c8c3a0ca17eff723d7-Paper-Conference.pdf.

[3] **Zander W. Blasingame** and Chen Liu. "Fast-DiM: Towards Fast Diffusion Morphs". In: *IEEE Security & Privacy* 22.4 (2024), pp. 103–114. DOI: 10.1109/MSEC.2024.3410112.

[4] **Zander W. Blasingame** and Chen Liu. "Greedy-DiM: Greedy Algorithms for Unreasonably Effective Face Morphs". In: *2024 IEEE International Joint Conference on Biometrics (IJCB)*. [**Spotlight**]. 2024, pp. 1–11. DOI: 10.1109/IJCB62174.2024.10744517.

[5] **Zander W. Blasingame** and Chen Liu. "Leveraging Diffusion for Strong and High Quality Face Morphing Attacks". In: *IEEE Transactions on Biometrics, Behavior, and Identity Science* 6.1 (2024), pp. 118–131. DOI: 10.1109/TBIOM.2024.3349857.

[6] Richard E. Neddo, **Zander W. Blasingame**, and Chen Liu. "The Impact of Print-Scanning in Heterogeneous Morph Evaluation Scenarios". In: *2024 IEEE International Joint Conference on Biometrics (IJCB)*. 2024, pp. 1–10. DOI: 10.1109/IJCB62174.2024.10744441.

[7] Chutitep Woralert, Chen Liu, and **Zander Blasingame**. "Towards Effective Machine Learning Models for Ransomware Detection via Low-Level Hardware Information". In: *Proceedings of the 13th International Workshop on Hardware and Architectural Support for Security and Privacy*. HASP '24. Association for Computing Machinery, 2024, pp. 10–18. ISBN: 9798400712210. DOI: 10.1145/3696843.3696847. URL: https://doi.org/10.1145/3696843.3696847.

[8] Chutitep Woralert, Chen Liu, and **Zander Blasingame**. "HARD-Lite: A Lightweight Hardware Anomaly Realtime Detection Framework Targeting Ransomware". In: *IEEE Transactions on Circuits and Systems I: Regular Papers* 70.12 (2023), pp. 5036–5047. DOI: 10.1109/TCSI.2023.3299532.

[9] Chutitep Woralert, Chen Liu, **Zander Blasingame**, and Zhiliu Yang. "A Comparison of One-class and Two-class Models for Ransomware Detection via Low-level Hardware Information". In: *2023 Asian Hardware Oriented Security and Trust Symposium (AsianHOST)*. [**Oral**]. 2023, pp. 1–6. DOI: 10.1109/AsianHOST59942.2023.10409333.

[10] Chutitep Woralert, **Zander Blasingame**, and Chen Liu. "HARD-Lite: A Lightweight Hardware Anomaly Realtime Detection Framework Targeting Ransomware". In: *2022 Asian Hardware Oriented Security and Trust Symposium (AsianHOST)* (2022). [**Best Paper Nominee**].

[11] **Zander Blasingame** and Chen Liu. "Leveraging Adversarial Learning for the Detection of Morphing Attacks". In: *2021 IEEE International Joint Conference on Biometrics (IJCB)* (2021). [**Oral**], pp. 1–8. DOI: 10.1109/IJCB52358.2021.9484383.

[12] **Zander Blasingame**, Chen Liu, and Xin Yao. "Feature Creation Towards the Detection of Non-control-Flow Hijacking Attacks". In: *Artificial Neural Networks and Machine Learning – ICANN 2021*. Ed. by Igor Farkaš, Paolo Masulli, Sebastian Otte, and Stefan Wermter. [**Oral**]. Cham: Springer International Publishing, 2021, pp. 153–164. ISBN: 978-3-030-86362-3.

[13] Gildo Torres, Zhiliu Yang, **Zander Blasingame**, James Bruska, and Chen Liu. "Detecting Non-Control-Flow Hijacking Attacks Using Contextual Execution Information". In: *Proceedings of the 8th International Workshop on Hardware and Architectural Support for Security and Privacy*. HASP '19. Phoenix, AZ, USA: Association for Computing Machinery, 2019. ISBN: 9781450372268. DOI: 10.1145/3337167.3337168. URL: https://doi.org/10.1145/3337167.3337168.

[14] Chen Liu, Zhiliu Yang, **Zander Blasingame**, Gildo Torres, and James Bruska. "Detecting Data Exploits Using Low-Level Hardware Information: A Short Time Series Approach". In: *Proceedings of the First Workshop on Radical and Experiential Security*. RESEC '18. Incheon, Republic of Korea: Association for Computing Machinery, 2018, pp. 41–47. ISBN: 9781450357579. DOI: 10.1145/3203422.3203433. URL: https://doi.org/10.1145/3203422.3203433.

Workshop Presentations & Posters

[1] **Zander W. Blasingame** and Chen Liu. "Rex: A Family of Reversible Exponential (Stochastic) Runge-Kutta Solvers". In: *ICLR 2026 2nd Workshop on Deep Generative Model in Machine Learning: Theory, Principle and Efficacy*. 2026. URL: https://openreview.net/forum?id=rHIUKCqPBZ.

[2] Richard E. Neddo, Emmanuel A. Atindama, **Zander W. Blasingame**, and Chen Liu. "LoRA as a Flexible Framework for Securing Large Vision Systems". In: *ICLR 2026 Workshop on Principled Design for Trustworthy AI - Interpretability, Robustness, and Safety across Modalities*. 2026. URL: https://openreview.net/forum?id=mmM7CVfzVw.

[3] Romeo Passaro, **Zander W. Blasingame**, Michael M. Bronstein, and Alexander Tong. "Stochastic Few-step Models". In: *ICLR 2026 2nd Workshop on Deep Generative Model in Machine Learning: Theory, Principle and Efficacy*. 2026. URL: https://openreview.net/forum?id=nmczKNW73P.

[4] **Zander W. Blasingame** and Chen Liu. "A Reversible Solver for Diffusion SDEs". In: *ICLR 2025 Workshop on Deep Generative Model in Machine Learning: Theory, Principle and Efficacy*. 2025. URL: https://openreview.net/forum?id=0gEFLVUL6n.

[5] **Zander W. Blasingame** and Chen Liu. "Greed is Good: A Unifying Perspective on Guided Generation". In: *The Exploration in AI Today Workshop at ICML 2025*. 2025. URL: https://openreview.net/forum?id=cSU2jEzanw.

[6] **Zander W. Blasingame** and Chen Liu. "Greed is Good: Guided Generation from a Greedy Perspective". In: *ICLR 2025 Workshop on Frontiers in Probabilistic Inference: Learning meets Sampling*. 2025. URL: https://openreview.net/forum?id=o4yQzZ5qCW.

[7] Richard E. Neddo, Sean Willis, **Zander W. Blasingame**, and Chen Liu. "Adapting Pretrained Vision Transformers with LoRA Against Attack Vectors". In: *IEEE International Conference on Mobility: Operations, Services, and Technologies*. 2025.

[8] **Zander W. Blasingame** and Chen Liu. "AdjointDEIS: Efficient Gradients for Diffusion Models". In: *The 3rd New Frontiers in Adversarial Machine Learning Workshop (AdvML @ NeurIPS)*. Vancouver, Canada, Dec. 2024.

[9] **Zander W. Blasingame** and Chen Liu. "Leveraging Diffusion Models for Stronger Face Morphing Attacks". In: *European Association for Biometrics (EAB) and the Center for Identitification Technology Research (CITeR) Biometrics Workshop*. Martigny, Swizterland, Apr. 2023.

[10] Nasser Nasrabadi, Chen Liu, David Doermann, and **Zander W. Blasingame**. "Face Morph Generation and Attack Detection". In: *International Face Performance Conference*. [**Oral**]. Online, Nov. 2022.

[11] Chen Liu, **Zander W. Blasingame**, Jeremy Dawson, and Jacob Dameron. "Morph attack detection and mitigation projects". In: *International Face Performance Conference*. [**Oral**]. Online, Oct. 2020.

Invited Talks

Mila - Quebec Artificial Intelligence Institute — Jan 2025
AdjointDEIS: Efficient Gradients for Diffusion Models — *Montréal, Canada*

Transatlantic Dialogue on Presentation Attack Detection — Nov 2024
Diffusion Morphs (DiM): Diffusion is all you need for highly effective face morphs — *Washington, D.C.*

Journal Presentation - IEEE IJCB — Sep 2024
Diffusion Morphs (DiM): Leveraging Diffusion for Strong and High-Quality Face Morphing Attacks — *Buffalo, NY*

Idiap Research Institute — Jul 2024
Diffusion Morphs (DiM): The Power of Iterative Generative Models for Attacking FR Systems — *Martigny, Switzerland*

CITeR and DSA Webinar — Feb 2024
Diffusion for the Generation of Face Morphs — *Online*

Awards and Honors

NeurIPS Top Reviewer Award — 2025

AISTATS Best Reviewer Award — 2025

IJCB Doctoral Consortium — 2024

Clarkson Presidential Scholar — 2015 - 2018

Best Oral Presentation for Software Engineering - Clarkson RAPS — 2017

Research Funding

Explainable Image Quality with Transformer-based Models — 2023 - 2024
Funding from the Center for Identification Technology Research a NSF-IUCRC
Role: Helped draft grant proposal, primary student investigator
PIs: Chen Liu

Towards the Creation of a Large Dataset of High-Quality Face Morphs — 2021 - 2024
Funding from the Center for Identification Technology Research a NSF-IUCRC
Role: Helped draft Clarkson's portion of grant proposal, primary student investigator at Clarkson University
PIs: Chen Liu, Stephanie Schuckers, Xin Li, Jeremy Dawson, Nasser Nasrabadi, David Doermann, Srirangaraj Setlur, Siwei Lyu, Xiaoming Liu, Sébastian Marcel

Comparative Detection of Facial Image Manipulation Techniques — 2020 - 2022
Funding from the Center for Identification Technology Research a NSF-IUCRC
Role: Helped draft grant proposal, primary student investigator
PIs: Chen Liu

Adversarial Learning Based Approach Against Face Morphing Attacks — 2019 - 2021
Funding from the Center for Identification Technology Research a NSF-IUCRC
Role: Helped draft grant proposal, primary student investigator
PIs: Chen Liu

Projects

Efficient Gradients for Diffusion Models - AdjointDEIS — AdjointDEIS

- Efficient algorithms for backpropagation in diffusion models for guided generation and related tasks
- Tools Used: Python, PyTorch

Diffusion Morphs (DiM) — Greedy-DiM

- Developed a family of SOTA face morphing algorithms using diffusion models
- Tools Used: Python, PyTorch

Teaching

Griffiss Institute 2020 - 2024
Instructor *Rome, NY*

- Taught and co-designed the introduction to cybersecurity summer camp for advanced high school students

Clarkson University 2018 - 2021
Graduate Teaching Assistant *Potsdam, NY*

- Electrical Science (ES 250)
- Introduction to Digital Design (EE 264)
- Electrical and Computer Engineering Sophomore Lab (EE 211)

Clarkson University 2016 - 2018
Undergraduate Teaching Assistant *Potsdam, NY*

- Software System Architecture (EE 462)
- Differential Equations (MA 232)

Professional Services

Reviewer

ICLR	2026
ICML	2025-2026
NeurIPS	2024-2025
AISTATS	2025
NeurIPS Workshop: 2nd edition of Frontiers in Probabilistic Inference: Learning meets Sampling	2025
ICCV 2nd Workshop on the Challenge Of Out Of Label Hazards In Autonomous Driving	2025
ICLR Workshop on Deep Generative Model in Machine Learning: Theory, Principle and Efficacy	2025
ICLR Workshop on Frontiers in Probabilistic Inference	2025
IJCB Workshop on Face Morphing Attack and Detection Techniques	2024
IEEE ICPR	2024
IEEE MOST	2023-2024
PMAM	2022-2023
ICANN	2021
IEEE BTAS	2019

Organizer

IJCB Workshop on Face Morphing Attack and Detection Techniques 2024
(Student Organizer)

Skills

ML Architectures: Diffusion Models, Flow Matching, Neural Differential Equations, GANs, VAEs

Distributed Computing: Slurm, CUDA, ROCm, Linux

Languages: Python, C/C++, Java, Javascript, Bash, MATLAB, VHDL

Frameworks: TensorFlow, Pytorch, Numpy, Scipy, Matplotlib, Plotly, Pandas, Jax

Tools: Vim, Git, Docker, LaTeX

Audio: Over a decade of exeperience as a FOH engineer including theater productions and worship services

References

References available upon request